National University of Computer and Emerging Sciences
Islamabad Campus

Programming Fundamentals (CS1002)

Date:26th May, 2025

Final Exam

Total Time (Hrs.): 3
Total Marks: 100
Total Questions: 6

Course Instructor(s)
Engr. Muhammad Hammad

_____ Roll No
_____ Section
_____ Student Signature

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Attempt all the questions.

CLO # 03: Illustrate the use of pointers, structures and memory management in C++.
[70 Marks]

Instructions: This question paper has six questions and eleven pages. Attempt Q2 and Q3 on question paper and remaining questions on answer sheet on given page numbers.

Q1: **Produce** the output of each code snippet.

(Attempt on Answer Book Page no 2,3,4)
[40 Marks (5 each)]

Ser No	Code Snippet

a	`#include<iostream>`
	`using namespace std;`
	`struct student`
	`{`
	`int id;`
	`string name;`
	`};`
	`int main()`
	`{`
	`student s={10,"Ahmed"};`
	`cout<<"ID: "<<s.id+10<<" , "<<s.name<<endl;`
	
	`}`

b	<pre>#include<iostream>
	using namespace std;
	const int size =2;
	struct Position
	{
	float xpos[size];
	float ypos[size];
	
	};
	
	Position print(Position q[])
	{
	for(int i=0;i<size;i++)
	{
	for(int j=0;j<size;j++)
	{
	cout<<"["<<q[i].xpos[j]<<" , "<<q[i].ypos[j]<<"]";
	if(j!=size-1)
	cout<<" , ";
	}
	cout<<endl;
	}
	}
	int main()
	{
	Position P[size];
	float temp=0.5;
	for(int i=0;i<size;i++)
	{
	for(int j=0;j<size;j++)
	{
	P[i].xpos[j]=temp;
	P[i].ypos[j]=temp+1;
	temp=temp*2;
	}
	//temp=temp*2;
	}
	print(P);
	
	}</pre>

c	<pre>#include<iostream>
	using namespace std;
	struct car</pre>

	{
	int length;
	int width;
	};
	int main()
	{
	int area;
	car *c=nullptr;
	c=new car[3];
	c[0]={10,20};
	c[1]={5,10};
	c[2]={6,7};
	for(int i=0;i<3;i++)
	{
	cout<<"Area "<<i<<" is : "<<(c[i].length)*(c[i].width)<<endl;
	}
	delete[] c;
	}

d	#include<iostream>
	using namespace std;
	
	int main()
	{
	int *ptr=nullptr;
	int c=10;
	ptr=&c;
	cout<<"The value is : "<<*ptr<<endl;
	c=c+(*ptr);
	cout<<"The value is :"<<c<<endl;
	*ptr=c+c;
	cout<<"The value is :"<<*ptr<<endl;
	
	}

e	#include<iostream>
	using namespace std;
	int calculate(int m,int n, const int *arr,int size)
	{
	cout<<"X: "<<m<<" , Y:"<<n<<endl;
	cout<<endl;
	for(int i=0;i<size;i++)
	{
	cout<<*arr+i<<",";
	}
	cout<<endl<<endl;

	return 0;
	
	}
	int main()
	{
	const int size = 5;
	const int numbers1[size] = {10, 20, 30, 40, 50};
	int x,y;
	for(int i=0;i<3;i++)
	{
	x=numbers1[i];
	y=numbers1[size-i-1];
	calculate(x,y,numbers1,size);
	}
	return 0;
	}

f	#include<iostream>
	using namespace std;
	int main()
	{
	int a=10;
	int b=20;
	int *ptr=&a;
	int **dptr=&ptr;
	cout<<*ptr<<endl;
	cout<<**dptr<<endl;
	**dptr=*ptr+a+b;
	cout<<a<<" , "<<*ptr<<" , "<<**dptr+(*ptr)<<endl;
	}

g	#include<iostream>
	using namespace std;
	const int size=9;
	
	int main()
	{
	int arr[size];
	int a2[]={10,20,30,40,50,60,70,80,90};
	int f=0;
	int l=size-1;
	int m,c=0;
	
	m=f+l/2;
	for(int i=m;i<=l;i++)
	{

```
                arr[i]=a2[c];
                c++;
        }
        for(int j=(m-1); j>=0;j--)
        {
                arr[j]=a2[c];
                c++;
        }
        for(int k=0;k<size;k++)
        {
                cout<<arr[k]<<" , ";
        }
}
```

h

```
#include<iostream>
using namespace std;
int main()
{
        bool choice=true;
        for (int i=15;i>=0;i--)
        {
                if(i==6)
                {
                        choice=false;
                }
                else if(i%4==0)
                {
                        continue;
                }

                else if(choice)
                {
                        cout<<"This is "<<i<<"th Loop Iteration"<<endl;
                        cout<<"_____"<<endl;
                }
                else
                break;

        }
}
```

Q2: **Apply** your knowledge of C++ to complete the structure of the given code by writing appropriate statements in the blank areas. Your code should produce the following output on screen (console). **Solve on question paper.** **[10 Marks]**

Output

```
Weekly Salary of Employee having ID: 101 (Ali) is Rs. 800
Weekly Salary of Employee having ID: 102 (Jameel) is Rs. 2000
Weekly Salary of Employee having ID: 103 (Sohail) is Rs. 1600
Weekly Salary of Employee having ID: 104 (Nasir) is Rs. 2800
Weekly Salary of Employee having ID: 105 (Kamran) is Rs. 1000
```

```
#include <iostream>
using namespace std;

struct Employee {
    string name;
    int id;
    int wph;      // wage per hour
    float salary;
};

//Insert function to calculate weekly salary of each employee using pointer to structure
```

//Insert function to print employee info (according to given output) using pointer to structure

```
int main()
 {
   const int size = 5;
```

Declare the Structure Variable

```
   =  {{"Ali", 101, 20},
       {"Jameel", 102, 50},
       {"Sohail", 103, 40},
       {"Nasir", 104, 70},
       {"Kamran", 105, 25}};

   int th = 8;                          // total hours per day
   int td = 5;                          // total days in a week
   calculate_salary(P, size, th, td);
   Print(P, size);
   return 0;
}
```

Q3: **Apply** your knowledge of C++ to complete the structure of the given code by writing appropriate statements in the blank areas. Your code should produce the following output on screen (console). **Solve on question paper.** **[10 Marks]**

.

```
Enter number of terms: 10
Fibonacci series: 0 1 1 2 3 5 8 13 21 34
```

```
#include <iostream>
using namespace std;

//Function to print to Fibonacci sequence using pointers only


//Function to generate Fibonacci series using pointers only


```

```
int main() {
    int n;

    cout << "Enter number of terms: ";
    cin >> n;

    if (n <= 0) {
        cout << "Please enter a positive integer." << endl;
        return 1;
    }

    int *fib =

    generateFibonacci(fib, n);    // function to generate the Fibonacci sequence using pointer only
    printFibonacci(fib, n);       // function to Print the Fibonacci sequence using pointer only

    //insert code to free the allocated memory

return 0;
}
```

Insert code to dynamically allocate memory of n size

Q4: Mr. Amir has 5 children i.e. 3 sons and 2 daughters. Each child has the following information:

(Attempt on Answer Book Page no 5,6,7)
[10 Marks]

- Name (string)
- Age (integer)
- Address, which includes:
- House number (integer)
- Street name (string)
- City (string)

Mr. Amir wants to maintain a record of all his children. You are required to:

1. **Apply** appropriate nested structures to represent each child and their address.

2. Store the information of all 5 children in an array of structures.
3. Sort the array in descending order of age.
4. Display the sorted information of all children in a neat format.

Sample Output

```
Mr. Amir's Children (Sorted by Descending Age):
==============================================
Name: Ahmed
Age: 22
Address: 303, Third Road, Islamabad
-----------------------------
Name: Bilal
Age: 21
Address: 505, Fifth Boulevard, Multan
-----------------------------
Name: Ali
Age: 20
Address: 101, Main Street, Lahore
-----------------------------
Name: Ayesha
Age: 19
Address: 404, Fourth Lane, Faisalabad
-----------------------------
Name: Sara
Age: 18
Address: 202, Second Avenue, Karachi
-----------------------------
```

CLO # 02: To use file handling in C++ and Develop modular code using functions to efficiently solve programming problems. [15 Marks]

Q5: Demonstrate C++ functions Pair_finder() and Print_Pair (). Take two arrays (a and b) as input from user**.** Pair_finder() should find the pairs of matching elements of a with b. Print_pair function should display the pairs value and its indices in both a and b. A sample output is given for reference. **(Attempt on Answer Book Page no 8,9)**

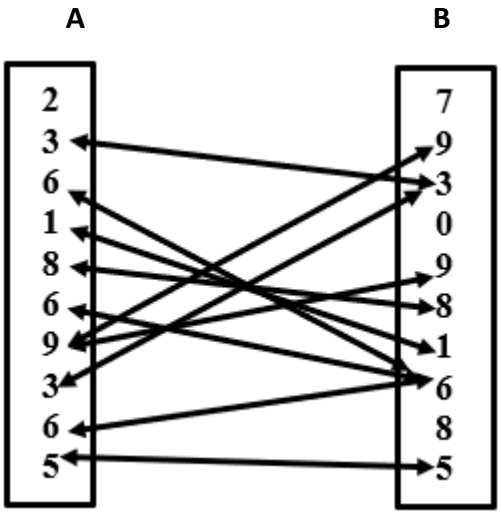
[15 Marks]

```
Enter number of elements in first array: 10
Enter elements of first array: 2,3,6,1,8,6,9,3,6,5
Enter number of elements in second array: 10
Enter elements of second
array:7,9,3,0,9,8,1,6,8,5
```

Common elements and their indices:

```
Value 3 found at a[1] and b[2]
Value 6 found at a[2] and b[7]
Value 1 found at a[3] and b[6]
Value 8 found at a[4] and b[5]
Value 8 found at a[4] and b[8]
Value 6 found at a[5] and b[7]
Value 9 found at a[6] and b[1]
Value 9 found at a[6] and b[4]
Value 3 found at a[7] and b[2]
Value 6 found at a[8] and b[7]
Value 5 found at a[9] and b[9]
```

The following diagram defines the problem



CLO # 01: To understand the concept and use of iterative structures, arrays and string operations.
[15 Marks]

Q6. **Demonstrate** a C++ Program which should extract **Main diagonal** and **anti-diagonal** from a 3x3 matrix (declare the matrix). Program should calculate the sum of both diagonal entries and should also calculate the transpose of matrix. A sample output is given for reference.

(Attempt on Answer Book Page no 10,11)
[15 Marks]

Sample Output:

Original Matrix:

3	4	6
7	8	9
10	13	6

Main Diagonal: 3 8 6
Sum of Main Diagonal: **17**
Anti-Diagonal: 6 8 10
Sum of Anti-Diagonal: **24**

Transpose of Matrix:

3	7	10
4	8	13
6	9	6

……*Best of Luck*…….